Chapter 5

5.1. The differential equation of a known continuous time LTI system is

$$y''(t)+4y'(t)+3y(t)=2x'(t)+x(t)$$

Try to find the frequency response $H(\mathrm{j}\omega)$ and impulse response $h(t)$ of the system.

5.2.We know a conscious time LTI system with input

$x(t)=\mathrm{Sa}(t)\cos(2t)+\mathrm{Sa}(t)\cos(4t)$, $-\infty<t<\infty$ and output

$y_{zs}(t)=\mathrm{Sa}(t)\cos(3t)$, $-\infty<t<\infty$,try to determine the frequency response of the system

5.3. We know the frequency response of a continuous time LTI system shown in the figure 5-1. Input signal is $x(t)=5+3\cos(2t)+\cos(4t),\quad -\infty<t<\infty$,try to determine steady state response of system *y(t).*

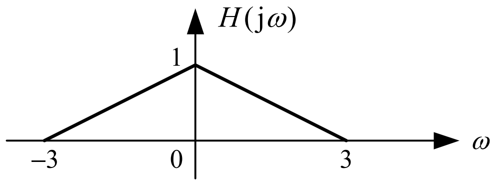


Figure.5-1

5.4.We know the differential equations describing the continuous time LTI system are as follows. Try to find the zero-state response of the system $y_{zs}(t)$.

(1) $y'(t)+3y(t)=2x(t),\quad t>0$, $x(t)=\mathrm{e}^{-4t}u(t)$

(2) $y''(t)+5y'(t)+6y(t)=3x'(t)+5x(t),\quad t>0$, $x(t)=u(t)$

5.5. Determine whether the continuous time LTI system is a undistorted transmission system, and calculate the impulse response of the system.

(1) $H(\mathrm{j}\omega)=2$ (2) $H(\mathrm{j}\omega)=3\mathrm{e}^{-\mathrm{j}2\pi t_0\omega}$

(3) $H(\mathrm{j}\omega)=\begin{cases}2\mathrm{e}^{-\mathrm{j}2\pi t_0\omega} & |\omega|\le\omega_c\\ 0 & others\end{cases}$ (4) $H(\mathrm{j}\omega)=3\mathrm{e}^{-\mathrm{j}(2\pi t_0\omega+\pi)}$

5.6. It is known that the amplitude response and phase response of a high-peaker are shown in figure 5-2.Among $\omega_c=80\pi$ *rad/s*

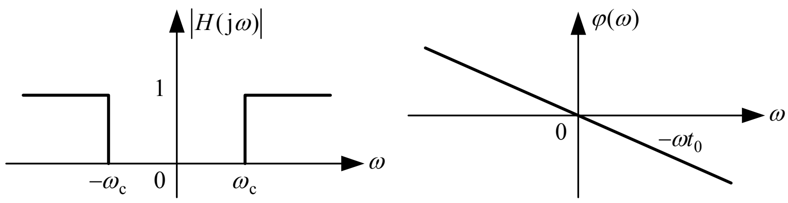


Figure.5-2

(1) Calculate the unit impulse response $h(t)$ of the system.

(2) If the input signal is $x(t)=1+0.5\cos(60\pi t)+0.2\cos(120\pi t)$,Try to determine the steady state response of the system

5.7. Frequency response of an ideal 90° phase shifter $H(\mathrm{j}\omega)$ is defined as

$$H(\mathrm{j}\omega)=\begin{cases}\mathrm{e}^{-\mathrm{j}\frac{\pi}{2}} & \omega>0\\ \mathrm{e}^{\mathrm{j}\frac{\pi}{2}} & \omega<0\end{cases}$$

(1)Try to determine the unit impulse response of the system.

(2)If the input is $x(t)=\sin(\omega_0 t)$, $-\infty<t<\infty$, finding the output $y_{zs}(t)$ of the system.

(3) Try to find the output $y_{zs}(t)$ of the system to any input x (T).

5.8. In the system shown in figure 5-3, we know the frequency spectrum $X(\mathrm{j}\omega)$ of the input signal $x(t)$. Try to analyze signals of A, B, C and D and specturn of y(t),plot the spectrum and find the relationship between $y(t)$ and $x(t)$.

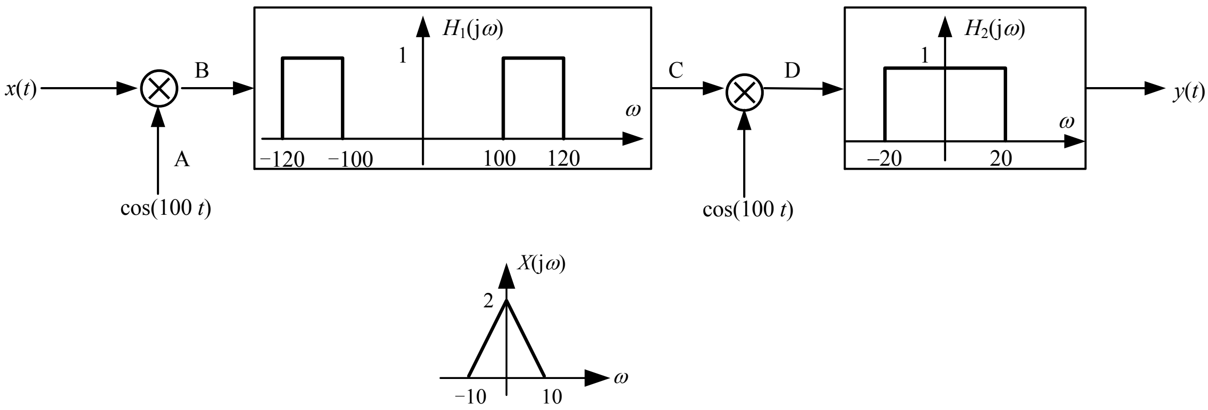
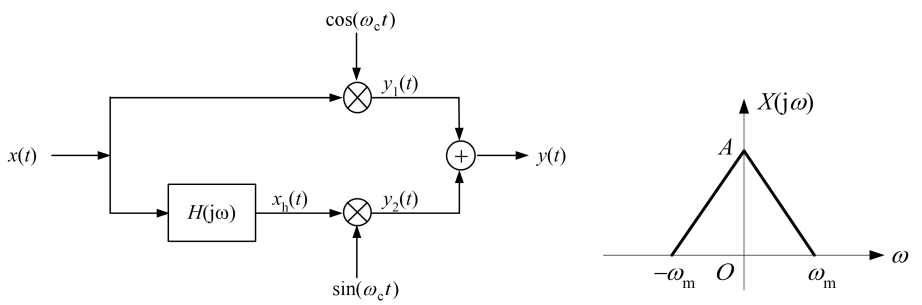


Figure 5-3

5.9. Figure 5-4(a) shows the block diagram of the single sideband modulation using Hilbert transform,we know the spectrum *X (jω)* of input signal *x (t)* is shown as the figure 5-3(b). When

H(jω) has the following conditions, try to plot spectrums of $y_1(t)$、$y_2(t)$ and $y(t)$.

(1) $H(\mathrm{j}\omega)=-\mathrm{j}\cdot\mathrm{sgn}(\omega)$ (2) $H(\mathrm{j}\omega)=\mathrm{j}\cdot\mathrm{sgn}(\omega)$



(a) (b)

Figure.5-4

5.10 A two amplitude modulation system shown in figure 5-5, the frequency band of the modulated signal $x(t)$ is 0.3～3.4kHz, first carrier signal frequency $f_1=100\mathrm{kHz}$, second carrier signal frequency $f_2=10\mathrm{MHz}$, we want to preserve the upper sideband spectrum of the bandpass filter's output signal $s_2(t)$. How much is the effective passband frequency range of two bandpass filters?

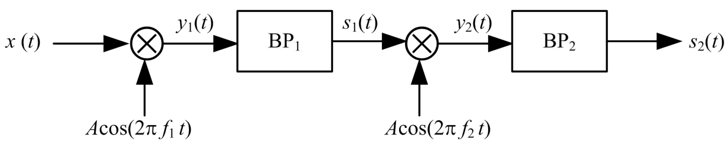


Figure.5-5

5.11. It is known that the difference equation of discrete-time LTI system is as follows. We try to get the frequency response $H(\mathrm{e}^{\mathrm{j}\Omega})$ and unit impulse response $h[k]$ of the system.

(1) $y[k] = x[k] + 2x[k-1] + x[k-2]$

(2) $6y[k] + 5y[k-1] + y[k-2] = x[k] + x[k-1]$

5.12.It is known the unit pulse response of the discrete LTI system is $h[k]=(0.5)^k u[k]$, calculate the frequency response $H(\mathrm{e}^{\mathrm{j}\Omega})$ of the system. When the input signal of the system is $x[k]=\cos(\pi k/4)$, try to derive the steady-state response of the system.

5.13.It is known the frequency response of a discrete band-pass filter is shown in Figure 5-6,try to find unit impulse response $h[k]$ of the filter. If the input of the filter is $x[k] = \sin(0.2\pi k) + 2\sin(0.5\pi k) + \sin(0.8\pi k)$, try to find the output $y[k]$ of the filter.

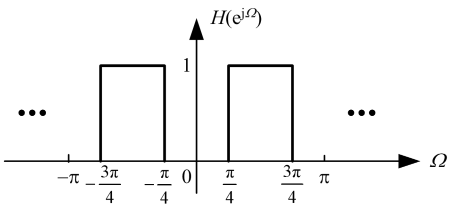


Figure.5-6